

Basic Probability, Conditional Probability, and Bayes' Theorem Questions

Basic Probability - Coin Toss

A coin is tossed once. What is the probability of getting a head?

Basic Probability - Dice Roll

A die is rolled once. What is the probability of getting a number greater than 4?

Conditional Probability - Coin Toss

A coin is tossed twice. What is the probability of getting heads on the first toss given that the second toss is a tail?

Conditional Probability - Card Draw

A card is drawn from a standard deck. Given that the card is red, what is the probability that it is a heart?

Conditional Probability - Dice Roll

A die is rolled. Given that the number is even, what is the probability it is a 2?

Bayes' Theorem - Coin Bag Example

There are two bags:

- Bag 1 contains 1 fair coin
- Bag 2 contains 1 double-headed coin

One bag is chosen at random, and a coin is tossed. The result is head. What is the probability that the coin was from Bag 2?

Bayes' Theorem - Medical Test

A disease affects 1 in 1000 people. A test for the disease is 99% accurate. If a person tests positive, what is the probability they actually have the disease?

Bayes' Theorem - Weather Prediction

A weather forecast says there is a 70% chance of rain. If it rains, there is a 90% chance the forecast was correct. If it doesn't rain, there's a 20% chance the forecast was wrong. Given that it rained, what is the probability that the forecast was correct?

Bayes' Theorem - Student Test Result

70% of students prepare for an exam. Among prepared students, 90% pass. Among unprepared students, only 30% pass. If a student passes, what is the probability that they had prepared?

Bonus: Conditional Probability from Joint Probability

In a class, 60% are boys and 40% are girls. 70% of boys and 50% of girls like math. If a student is randomly selected and is known to like math, what is the probability the student is a boy?